Palo Alto Firewall
App-ID Based Application Control

This report documents the step-by-step configuration and verification of application blocking (Facebook and YouTube) using Palo Alto Firewall (PA-VM). The objective of this lab was to restrict specific applications from a trusted internal host and verify the enforcement through traffic logs.

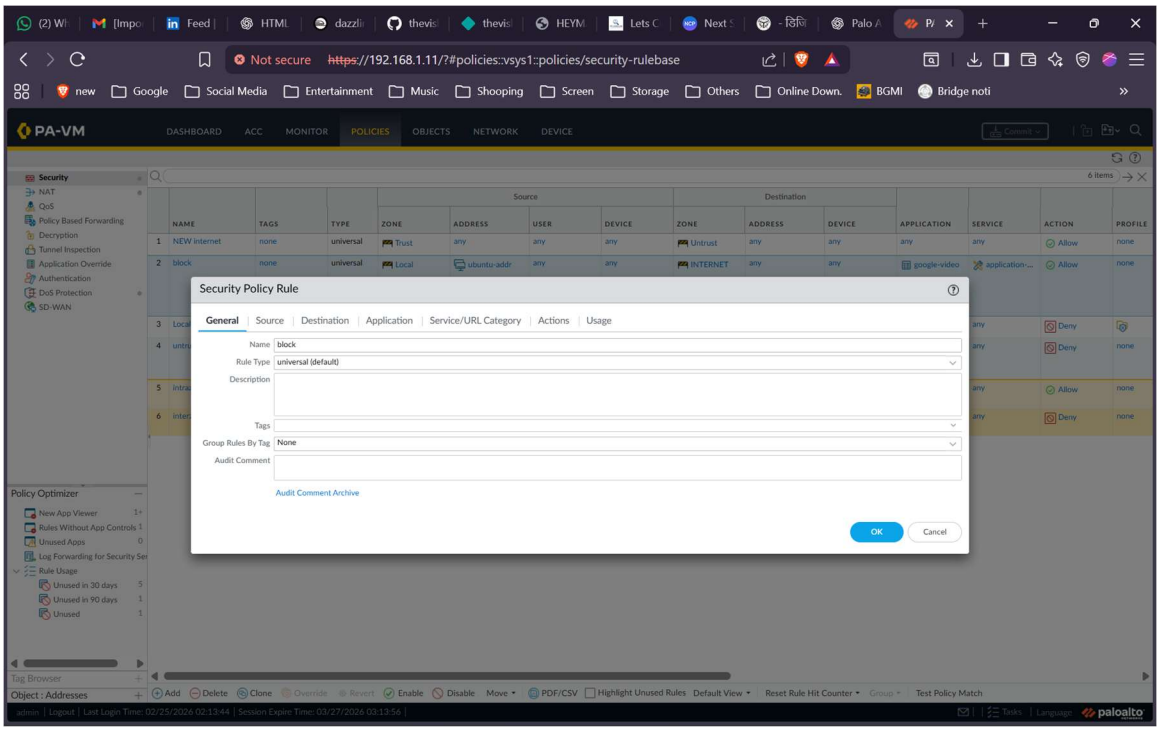
Network Topology Diagram

Test Environment Details

Firewall: Palo Alto PA-VM (ESX 11.2.0)
Client OS: Ubuntu 24.04 LTS
Virtualization Platform: VMware Workstation
Source Zone: Trust
Destination Zone: Untrust
Blocked Applications: Facebook, YouTube and related base applications

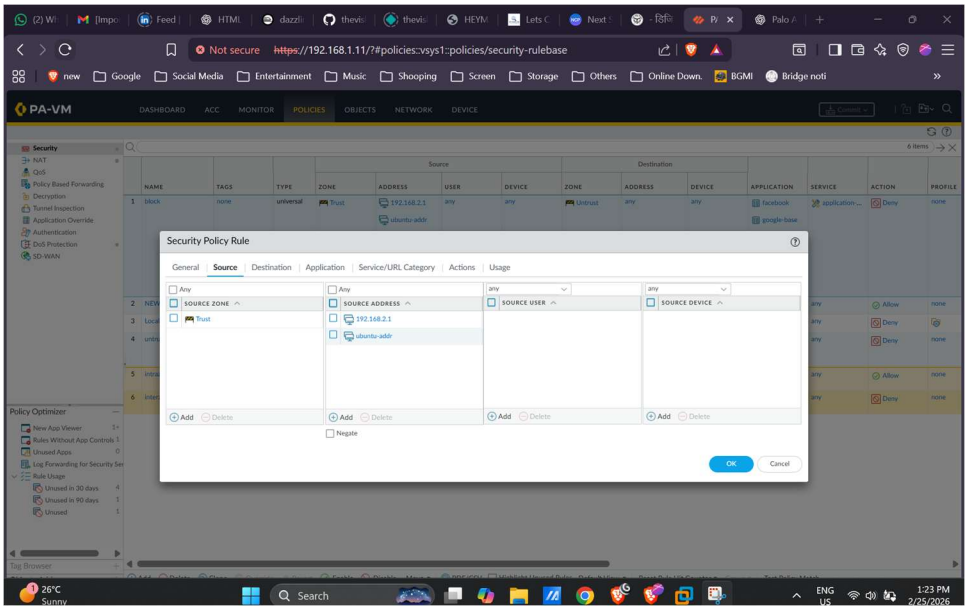
Step 1: Create Security Policy Rule

Navigate to Policies → Security and create a new rule named 'block'. Set the rule type as Universal and provide appropriate description if required.



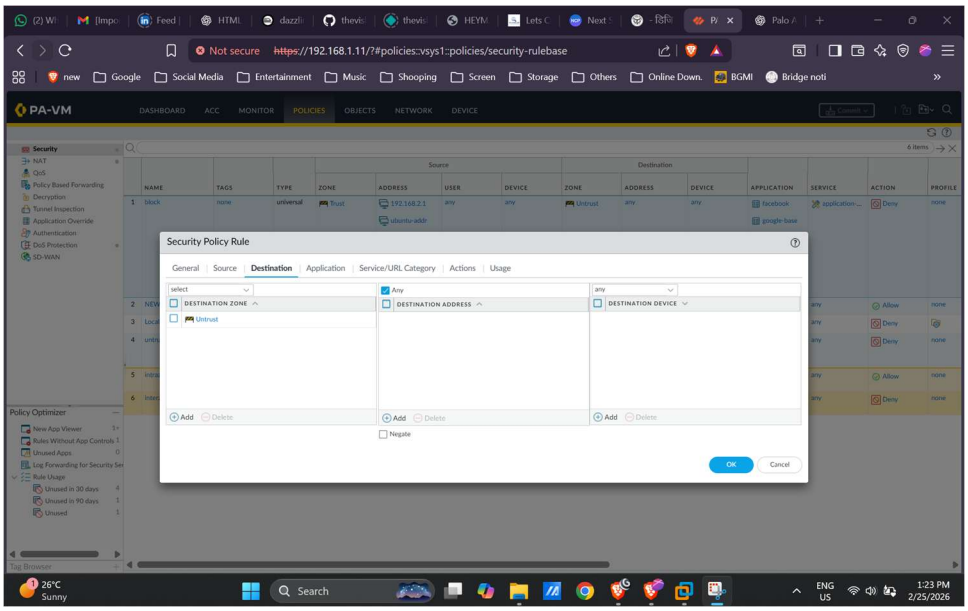
Step 2: Configure Source Settings

Under the Source tab, select Zone as Trust and specify the source address (192.168.2.1 / ubuntu-addr). This ensures the rule applies only to the intended host.



Step 3: Configure Destination Settings

Under the Destination tab, select Destination Zone as Untrust. Keep destination address as Any to block the selected applications externally.

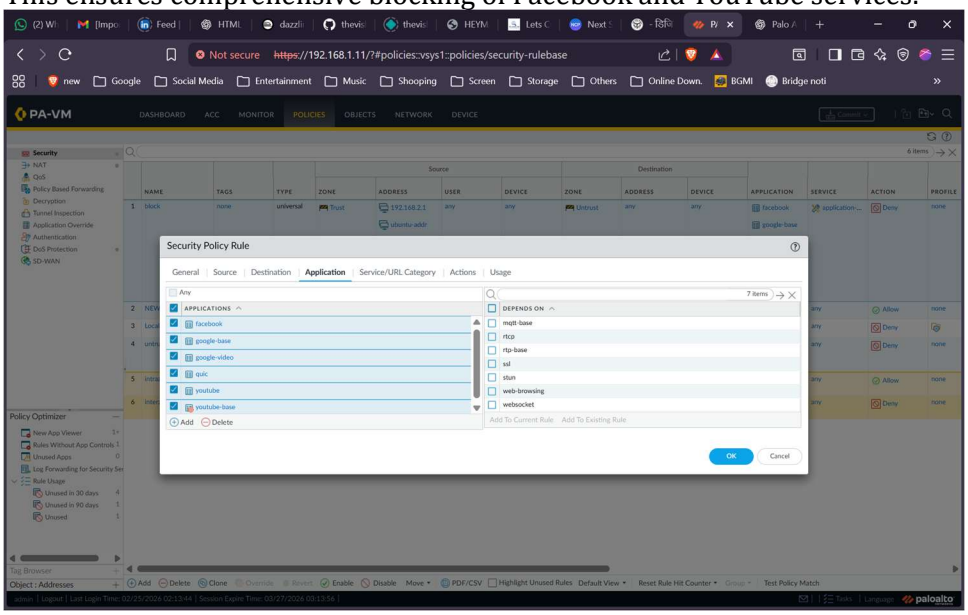


Step 4: Configure Applications

In the Application tab, add the following applications:

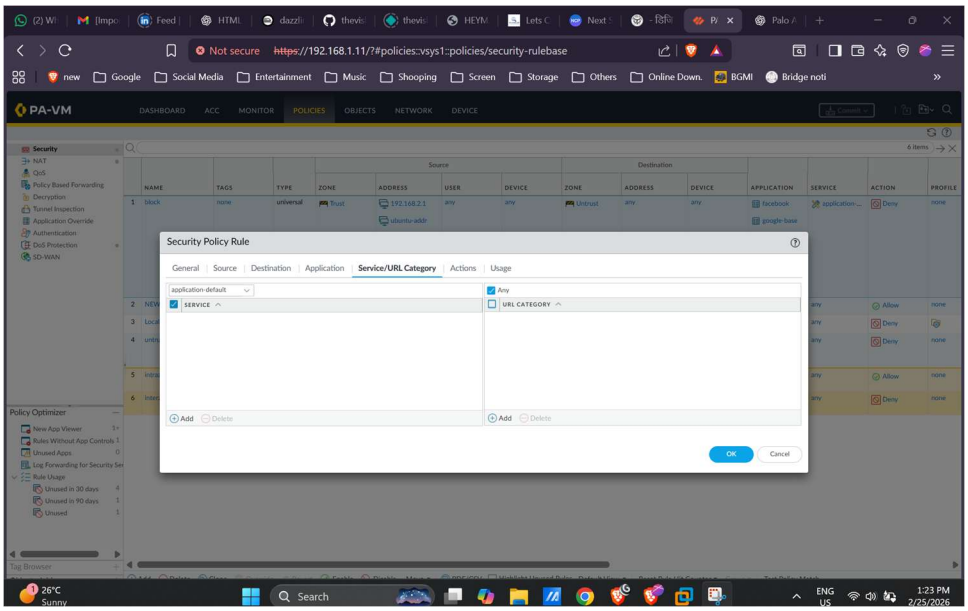
- facebook
- google-base
- google-video
- quic
- youtube
- youtube-base

This ensures comprehensive blocking of Facebook and YouTube services.



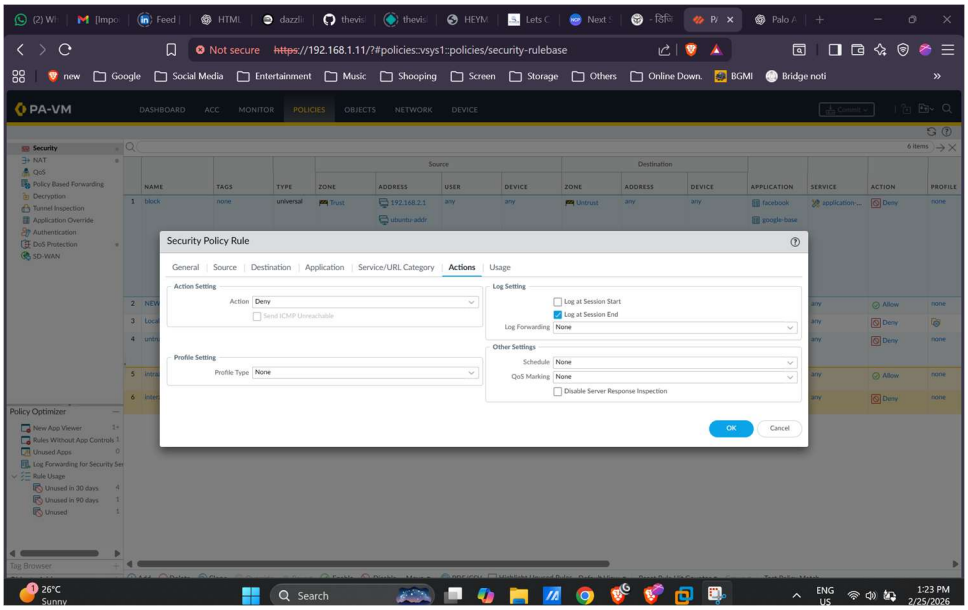
Step 5: Configure Service / URL Category

Under Service/URL Category, keep Service as application-default. This allows the firewall to match applications on their default ports.



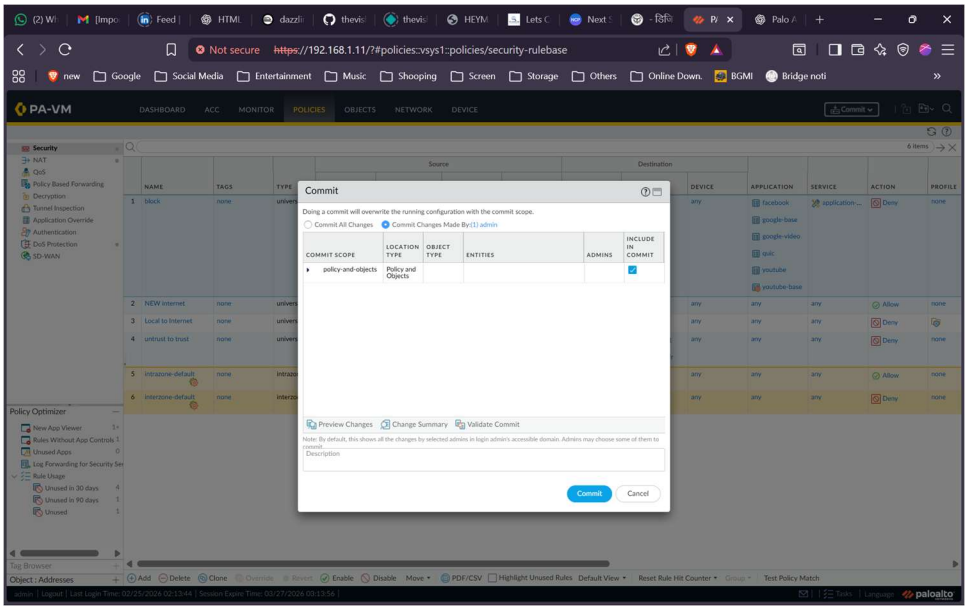
Step 6: Configure Action

In the Actions tab, set the Action to Deny and enable logging at session end to monitor blocked traffic in logs.



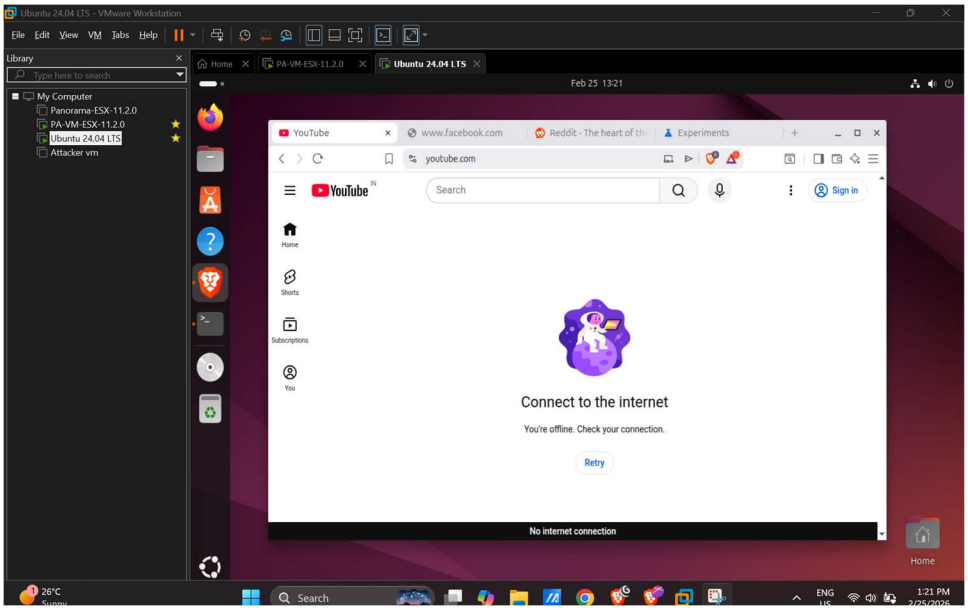
Step 7: Commit Configuration

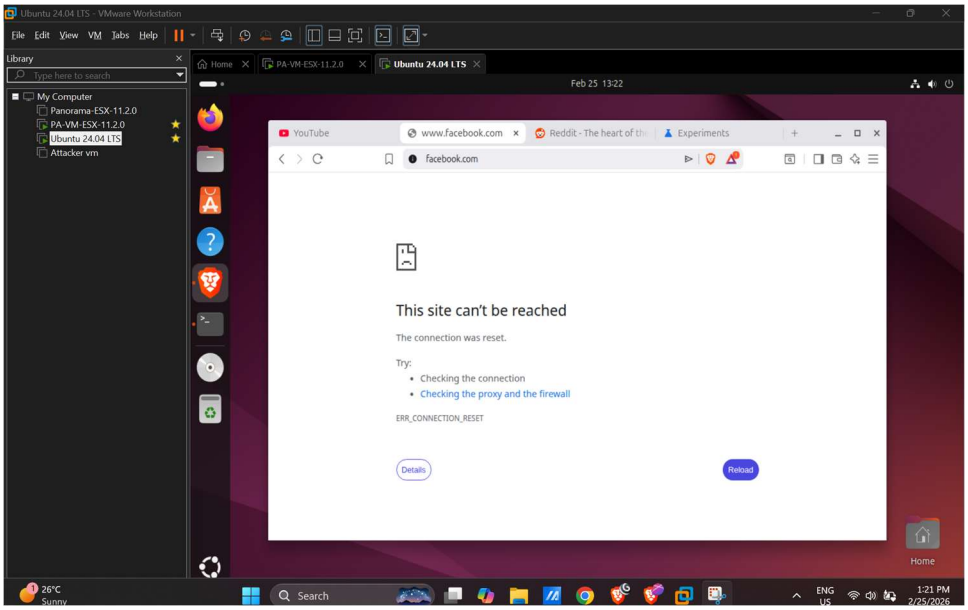
After configuring the rule, click Commit to apply the changes to the running configuration.



Step 8: Verification from Client Machine

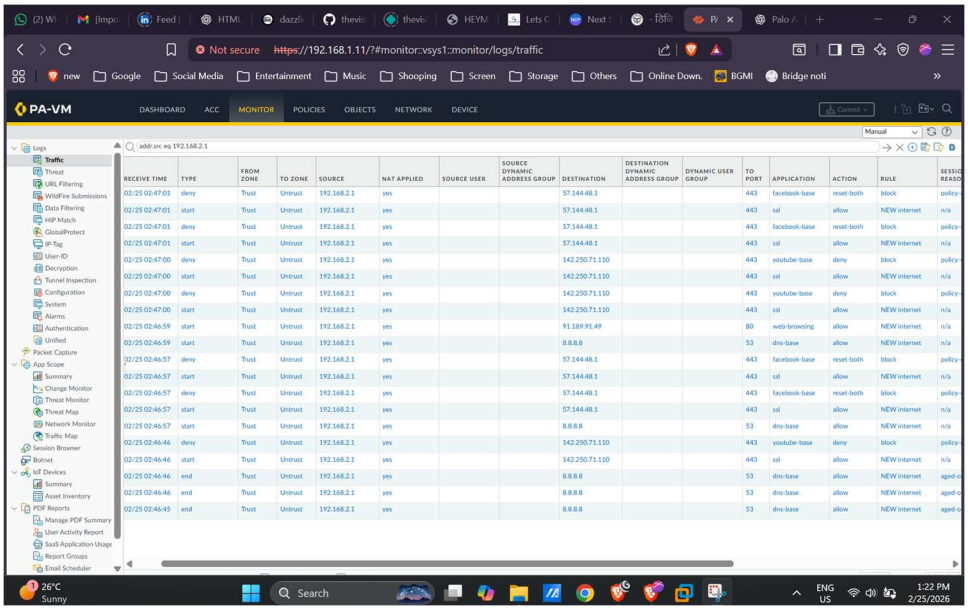
On the Ubuntu client machine, attempts to access YouTube and Facebook result in connection errors, confirming the rule enforcement.





Step 9: Traffic Log Verification

Navigate to Monitor → Traffic to verify blocked sessions. Logs show action as deny/reset-both with rule name 'block', confirming policy enforcement.



Conclusion

The lab successfully demonstrated application-based traffic control using Palo Alto Firewall. By implementing an App-ID based security policy, specific applications (Facebook and YouTube) were effectively blocked for a defined source host. Traffic logs confirmed that sessions were denied and reset as configured. This validates the effectiveness of Palo Alto's application-layer filtering capabilities.

- Vishesh Dixit